

0.8 Env Test

Intro.

Theorem 0.8.1 — Test Theorem

Let $A \in M_n(F)$. Then $\operatorname{tr}(A^\top) = \operatorname{tr} A$ and $\langle \mathbf{u}, \mathbf{v} \rangle, [T]_{\mathcal{B}}^{\mathcal{C}}, [\mathbf{v}]_{\mathcal{B}}, \operatorname{spec}(T)$.

Idea. Transpose keeps the diagonal.

Proof. Let $A = (a_{ij})$. This shows it. ■

Warning. The converse fails.

Quick check. Is $\operatorname{tr}(AB) = \operatorname{tr}(BA)$?

Solution. Yes, by Theorem 0.8.1.

0.8.1 Exercises

0.8.1.1 A. Check your understanding

Exercise A1.

- (a) Define trace.
- (b) State it.

Solution.

- (a) Sum of diagonal. (b) Done.